AI & Machine Learning - Task 3

Model Validation, Overfitting Control & Hyperparameter Tuning

1. Load Dataset

The California Housing Dataset was loaded using Scikit-learn. Features and target were separated, standardized using StandardScaler, and split into training and testing sets using an 80:20 ratio.

```
...    MedInc  HouseAge  AveRooms  AveBedrms  Population  AveOccup  Latitude  \
   0  8.3252      41.0  6.984127   1.023810       322.0  2.555556     37.88
   1  8.3014      21.0  6.238137   0.971880      2401.0  2.109842     37.86
   2  7.2574      52.0  8.288136   1.073446       496.0  2.802260     37.85
   3  5.6431      52.0  5.817352   1.073059       558.0  2.547945     37.85
   4  3.8462      52.0  6.281853   1.081081       565.0  2.181467     37.85

      Longitude  HousePrice
   0    -122.23       4.526
   1    -122.22       3.585
   2    -122.24       3.521
   3    -122.25       3.413
   4    -122.25       3.422
```

2. Base Model Performance

Linear Regression, Ridge Regression, Decision Tree, and Random Forest models were trained. Their performance was evaluated using RMSE, R^2 Score, and 5-Fold Cross Validation.

```
...  ============================================================
     BASE MODEL PERFORMANCE
     ============================================================

     Linear Regression
     Train RMSE : 0.72
     Test RMSE  : 0.746
     CV RMSE    : 0.746
     Train R2   : 0.613
     Test R2    : 0.576

     Ridge Regression
     Train RMSE : 0.72
     Test RMSE  : 0.746
     CV RMSE    : 0.746
     Train R2   : 0.613
     Test R2    : 0.576

     Decision Tree
     Train RMSE : 0.0
     Test RMSE  : 0.703
     CV RMSE    : 0.896
     Train R2   : 1.0
     Test R2    : 0.623

     Random Forest
     Train RMSE : 0.188
     Test RMSE  : 0.505
     CV RMSE    : 0.651
     Train R2   : 0.974
     Test R2    : 0.805
```

3.Overfitting Analysis

Training and testing performance were compared to identify overfitting. Decision Tree showed overfitting, while Random Forest generalized better by combining multiple decision trees.

...

```
============================================================
OVERFITTING ANALYSIS
============================================================
Linear Regression
Good Generalization

Ridge Regression
Good Generalization

Decision Tree
Model is likely Overfitting

Random Forest
Model is likely Overfitting
```

4.Hyperparameter Tuning – Random Forest

GridSearchCV with 5-Fold Cross Validation optimized n_estimators, max_depth, min_samples_split, and min_samples_leaf to improve model performance and reduce overfitting.

```
Best Random Forest Parameters
{'max_depth': None, 'min_samples_leaf': 2, 'min_samples_split': 2,
'n_estimators': 200}
```

5.Tuned Model Performance

The optimized Random Forest achieved lower RMSE, higher R^2 Score, and better validation performance than the baseline models.

```
============================================================
TUNED MODEL PERFORMANCE
============================================================
Decision Tree Tuned
RMSE : 0.639
CV RMSE : 0.769
R2 : 0.688

Random Forest Tuned
RMSE : 0.504
CV RMSE : 0.649
R2 : 0.806
```

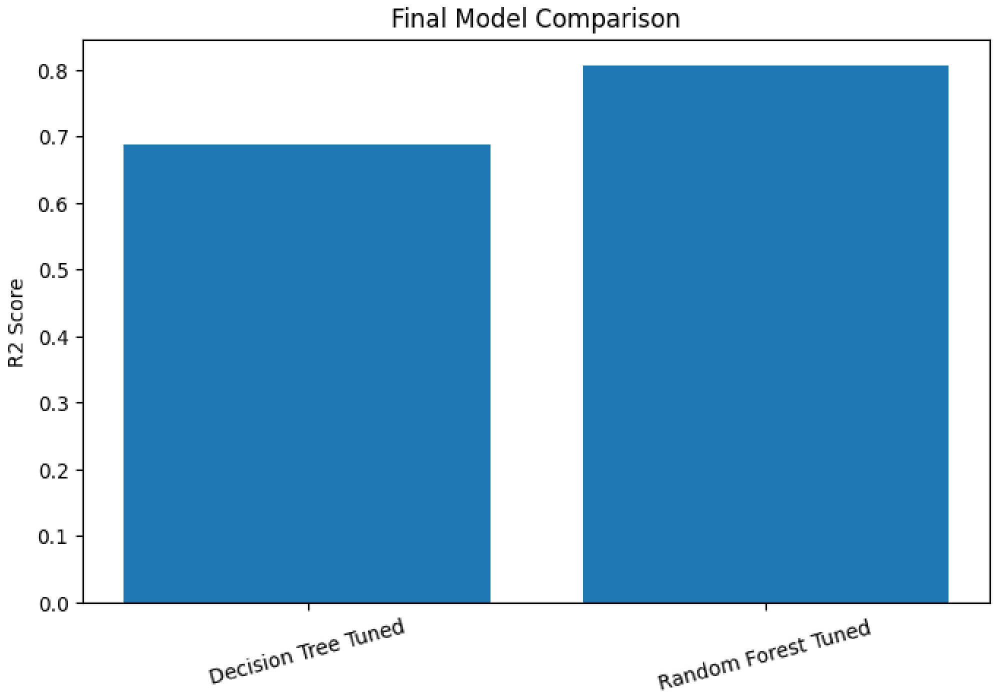
6.Model Comparison Table

Compare Linear Regression, Ridge Regression, Decision Tree, Tuned Decision Tree, Random Forest, and Tuned Random Forest using Test RMSE, Cross Validation RMSE, and R^2 Score. Replace placeholder values with notebook outputs.

```
==========================================================
FINAL MODEL COMPARISON
==========================================================
                Model  Test RMSE  Cross Validation RMSE  R2 Score
0  Decision Tree Tuned   0.638936               0.768669  0.688464
1  Random Forest Tuned   0.503949               0.649022  0.806194
```

7.Plot Comparison

Insert the model comparison bar chart generated by your notebook to visually compare model performance.



Conclusion

Model validation and hyperparameter tuning improved prediction accuracy and robustness. The Tuned Random Forest was selected as the final model because it achieved the best balance between accuracy and generalization.